

# Take home notes for: water kefir and ginger bug

During the workshop there can be a lot of new information to take in, don't worry! Here's a comprehensive guide to making water kefir and ginger bug.

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# Base Recipe 1: Water Kefir

Water Kefir is probably my favourite fermented soft drink (favourite fermented drink has to go to beer, sorry everything else). Unlike store bought soda's it can be low in sugar, high in probiotics and is flavoured and carbonated naturally.

It's also easy to make, not that fussy about conditions and the starter can be bought (or taken home from a workshop) and used straight away, no starter feeding needed.

Water kefir is fermented by the action of water kefir grains, little translucent clusters of bacteria and yeast living together. These are a form of SCOBY (Symbiotic Community Of Bacteria & Yeasts).

## **Equipment**

- A big clean jar or vessel (primary fermentation)
- A cloth or lid to cover
- A sieve or strainer
- Weighing scales
- A Kettle
- Pyrex jug
- Bottles (for the second ferment)

## **Ingredients**

- Water kefir grains
- Sugar (Demerara or brown is best)
- Water
- Flavourings (fruits or herbs)

## **Method**

To start with work out how much you want to make. The vessel, a big Kilner jar, that I usually use holds 2 litres comfortably so I'll go with the amounts for that but will also use percentages, so you can work it out for your fermenting vessel.

Boil a kettle. We only really need enough boiled water to dissolve the sugar, the rest will be cold water.

In a pyrex jug weigh out 120g sugar, this is 6% of the final 2 litres, so whatever 6% is for you. Add to this enough boiling water to dissolve it but not more than a quarter of the total, so 500ml here. Stir until fully dissolved. We now need to dilute this boiled sugar water with cold water, if we pour boiling water directly onto water kefir grains, they will die, so we need to bring the water temp down first.



I do this by putting 1.5l cold water into the fermenting vessel along with the water kefir grains, which which works as a buffer to the hot sugar liquid we pour in next, add your sugar water and if needed top up to the full 2litres or your total volume.

At this stage if you want to add flavourings you can do it now. (See [Flavouring ideas](#))

Place a loose fitting lid or cloth over your fermenting vessel and you're good to leave it on the side to ferment.

### **First Fermentation**

After a day or so you should start to notice small bubbles and rising to the surface, if you've added fruit that might start floating to the top. After about 2 days or when all the fruit is floating its ready for secondary fermentation. Remove the fruit, which should be floating at the top and strain the liquid into bottles. Any water kefir grains in the strainer can go back into the fermenting vessel for the next round of water kefir you make.

### **Second Fermentation**

Once in your bottles, seal them and leave at room temperature for 1-2 days. It will become fizzy in the bottle (See [Why are the drinks fizzy?](#)). Then move to the fridge over night before consuming.

### **Flavouring Ideas**

Simply add these things to your primary fermentation.

- **Strawberry, Basil and black pepper** - This might sound mad but it's delightful
- **Apple juice** - This you can add just by itself without sugar to the grains, it's very lively though so watch out for carbonation.
- **Mixed berries**, a small handful, crushed slightly to release their juice
- **Elderflower cordial**, Make up a fairly sweet squash and use this in place of the sugar water.

It's worth noting that some things (I'm looking at you black pepper) are really difficult to separate from the grains, if this becomes annoying, you might want to consider a 3 stage fermentation: starter, first and second ferment. Here you keep your water kefir grains regularly feed with sugar water in a smaller jar. When you want to make a batch up you use the liquid from this small jar in your big fermenting vessel and leave the grains behind. This is the same technique we use in Ginger Bug.

### **Looking after your grains**

Water kefir grains grow over time and will need dividing every so often, if you notice you have more than you started with, share the extra with a friend or start a second jar.



If you need a break from feeding them daily, you can slow them right down by storing them in fresh sugar water in the fridge for up to a couple of weeks.

### **Sugar and fruits**

In my experience water kefir is more happy when there are more nutrients. Brown and dark sugar will boost fermentation vigour as will the addition of fruits.

### **Is it Kefir?**

No! Kefir is a milk based fermentation that has a different set of microbes. I know it's confusing. Water kefir grains are also called Tibicos, which can be a useful distinction.

### **Why not kombucha?**

Within the world of home fermented tonic beverages, Kombucha currently seems to be king. I have made kombucha a lot over the years but these days I only make it for others for the simple fact that as it ages it can become incredibly sour, almost like drinking vinegar and I'm just not a fan of that as a drink. There is tons of information out there if you're interested and I would highly recommend you giving it a try if that's your bag, the process is very similar to making water kefir so the knowledge is transferable.

## **Base Recipe 2: Ginger Bug**

### **The Starter**

A ginger bug is a starter, that can then be used to make ginger beer, in the same way that people make a sourdough starter to then make sourdough bread.

It captures the natural yeasts and bacteria that live on the skin of fresh ginger and builds them up into an active little culture. Once established, you add a small amount to a sugary liquid to ferment it.

It takes about a week of daily feeding to get going, but once established it can be kept alive indefinitely with regular feeds, much like a sourdough starter.

My favourite thing about this is that if your starter dies for some reason. You can start a new one with easily available ingredients without having to buy a new starter.

### **Ingredients**

- Fresh ginger, unpeeled (organic where you can, it's more likely to have the microbes we need)
- Sugar
- Water



## **Method**

Finely grate or chop 2 tablespoons of ginger (skin on) and add it to a clean jar with 2 tablespoons of sugar and 1 cup of water. Stir well and cover loosely with a cloth or a screw on lid not fully tightened.

Each day after that, add another tablespoon of grated ginger and a tablespoon of sugar, stirring well each time. Keep the jar at room temperature, out of direct sunlight.

After 3-5 days you should start to see small bubbles rising when you stir, this is a sign the wild yeast and bacteria are multiplying. Keep feeding daily until it's reliably fizzy within a few hours of feeding, usually 5-7 days in total. If you're having trouble getting it going see [troubleshooting](#).

Your starter is now ready to use to make a probiotic ginger beer (or other kind of natural soda)

## **Using your ginger bug for ginger beer**

Instructions here are very similar to water kefir, we're using the same ratio of 6% sugar to water as our sugary liquid to ferment. However our "grains" here is our ginger bug starter. I'll be giving amounts for 2 litres, scale up or down as needed.

Once the starter is active, strain off the liquid (leaving the ginger pulp and bug behind in the jar to keep feeding with water and sugar) and save this liquid for the next step.

In a pan add 500ml water (25% of total) and 120g sugar (6% weight of total water) and bring to a simmer. Add a handful of chopped ginger, you can experiment with the exact amount given how fiery you like it. Simmer for 15 mins.

In your fermenting vessel add 1.5l cold water, strain in the ginger and sugar infusion, stir, add you ginger bug starter liquid, stir again and add a loose fitting lid or a cloth.

## **Fermentation**

You can follow the same fermentation steps as for water kefir. It is also worth noting that you don't have to make ginger beer. Your ginger bug starter can be used ferment any kind of sugary liquid base, why not try some of the flavour suggestions from above?

# **Fermentation Considerations**

## **Why are the drinks Fizzy?**

Both of these drinks use a kind of SCOBY (symbiotic community of bacteria and yeast) as a way to ferment the sugars in the base liquids. These microbes work collaboratively to consume the sugars and turn them into Carbon-



dioxide (Co<sub>2</sub>) which is what makes the drink fizzy. Whilst consuming the sugars the SCOBY also produces various acids, making the drink lightly tangy and trace amounts of alcohol, which with these sugar levels is about the same as non alcoholic beer.

## **Temperature**

Both of these ferments can be very temperature dependent and especially in our colder cumbrian climate, in some rooms they can be very sluggish to get going. Ideally you would want to keep them above 20°C. An airing cupboard or warm sunny window can work or even a very low power greenhouse heated seed mat.

## **Why a second ferment?**

The first ferment, whether it's water kefir or a ginger bug, is mostly about building up a healthy population of microbes and developing flavour. Because that stage happens in a jar covered with a cloth or loose fitting lid, most of the CO<sub>2</sub> the produced simply escapes into the air rather than carbonating the liquid.

Sealing the drink into a bottle for a second ferment traps that Co<sub>2</sub> instead of letting it escape, so it has nowhere to go but into the liquid, this is what gives both water kefir and ginger bug sodas their fizz. Once enough Co<sub>2</sub> has built up, moving the drink to the fridge helps the Co<sub>2</sub> dissolve into the liquid, making the drink retain more fizz.

## **A note on bottling**

WARNING - It is very important that you do not bottle water kefir or ginger bug drinks too early and then forget about them. Bottling early means more sugar in the bottle, which means more co<sub>2</sub> produced, which means more pressure built up. If this pressure exceeds the bottles pressure rating it could explode and send small pieces of glass flying through the air, not good. You should never use bottles for olive oil or wine as these are not pressure rated. If you are going to use glass use bottles designed to hold pressure.

Whilst I don't like using plastic in fermentation where possible, using plastic bottles when you start is a great way to see pressure building, as the bottle literally starts to bulge.

# **Troubleshooting**

## **Water kefir tastes flat, not fizzy**

The grains may need more time, a warmer spot, or a fresh feed of sugar water. If it's been weeks since they were properly active, give them a couple of batches to build their strength back up. It's also worth noting that water kefir likes a more complex sugar like Demerara or brown sugar and adding fruit to you ferment will help too.



### **Ginger bug not bubbling after a week**

Try a different ginger, an organic one or sometimes even just from a different super market. Some gingers are treated before sale to kill off microbes, which is exactly what we don't want.

A cooler kitchen will also slow things down considerably, try moving the jar somewhere warmer.

### **Second ferment bottles not carbonating**

Usually means not enough residual sugar or yeast activity, or the room is too cold. Add a touch more sugar or fruit juice before sealing, and give it a warmer spot.

### **Second ferment bottles overflowing when opened**

Too much sugar in the solution before bottling leading to over carbonation. Let your first fermentation have a little more time before bottling.